THE UNIVERSITY OF BRITISH COLUMBIA

# CONFINED DIFFUSION OF NANOPARTICLES USING iSCAT MICROSCOPY

by

Martin Jasinski

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# Abstract

The aim of this thesis is to characterize the full diffusion model of complex nanoparticles. To accomplish this, the 3D position of a nanoparticle(20-100 nm) must be measureable with a time resolution on the other of the diffusion length (ms). Interferometric light scattering microscopy (iSCAT) is used in tandem with convex-lens induced confinement (CLiC) to simultaneously confine particles, while obtaining a spatial signal.

This thesis developed a novel method of measuring axial position using iSCAT data. The method was applied to a hard-wall axial confinement potential. The system's ionic conditions, particle charge, and confinement length were varied to characterize the effect of the various variables.

The application of this thesis was directly applied to a more common, commericially assembled fluorescent microscope. The applications of the iScat confinement measurements allow for a direct correction to fluorescet microscopy calculations of diffusion in confinemnt. To accomplish this, an off-the-shelf optical system was designed. The system consists of a 3-color laser excitation with alternating laser excitation capabilities. The Nikon Ti2-Eclipse microscope chassis was used for this system in tandem with a 2-camera twincam system for dual color imaging. Imaging of nanoparticles was performed under various confinement regimes and corrections to the measuements was performed.

The results showed...

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# Chapter 1

# Introduction

## 1.1 Movivation

- What is the major field i am in? - what are the biological questions - how are they typically slved

- problems with the current solutions

## 1.2 Brownian Motion

Martin jasinsk is awesome

$$\vec{F} = ma \tag{1.1}$$

This is cool! MARTIN

JASINKSIS

### 1.2.1 Einstein's Diffusion

### 1.2.2 Langevin Dynamics

### 1.2.3 Electrostatics, Gravitation, and Potentials

## 1.3 Confined Diffusion Processes

### 1.3.1 From Langevin Dynamics to the Fokker-Plank

### 1.3.2 Geometric Confinement

### 1.3.3 Hydrodynamic Confinement

### 1.3.4 Example: Polystyrene Bead

## 1.4 In-Vivo Considerations

### 1.4.1 why does this matter for biological applications

## 1.5 Outline of Thesis Chapters

# Chapter 2

# Widefield backscattering iscat for SPT in 3d confinement tracking and analysis

## 2.1 Rayleigh scattering

## 2.2 Methods

## 2.3 Questions

# Chapter 3

# Conclusion